

python

Python is a high-level, interpreted, and object-oriented programming language developed by Guido van Rossum and released in 1991. It is one of the most popular programming languages in the world because of its simple syntax and readability. Python is easy to learn and is widely used by beginners as well as professional developers.

Python supports multiple programming paradigms such as procedural programming, object-oriented programming, and functional programming. It is platform-independent, which means Python programs can run on Windows, Linux, and macOS without major changes.

Python is widely used in web development, artificial intelligence, machine learning, data science, automation, cybersecurity, game development, desktop applications, and software testing. Popular frameworks like Django and Flask are used for web development in Python.

One of the biggest advantages of Python is its large collection of libraries and modules. Libraries such as NumPy, Pandas, Matplotlib, TensorFlow, and OpenCV help developers build powerful applications easily.

Python uses indentation instead of curly braces to define blocks of code. This makes the code clean and readable. Variables in Python do not require explicit datatype declaration because Python is dynamically typed.

Example:

```
x = 10
```

```
name = "Bharti"
```

Python provides various data types such as:

- Integer
- Float
- String
- List
- Tuple
- Dictionary

- Set

Python also supports functions, classes, inheritance, polymorphism, encapsulation, and other object-oriented concepts.

Loops in Python include:

- for loop
- while loop

Conditional statements include:

- if
- if-else
- nested if

Python has automatic memory management and garbage collection which makes development easier.

Advantages of Python:

1. Easy to learn
2. Simple syntax
3. Large community support
4. Huge libraries collection
5. Cross-platform support
6. Fast development

Disadvantages of Python:

1. Slower than compiled languages
2. High memory usage
3. Not ideal for mobile app development

Python is commonly used in modern technologies such as:

- Artificial Intelligence

- Machine Learning
- Data Analytics
- Web Applications
- Automation Tools
- Cloud Computing

Django is a high-level Python web framework used to develop secure and scalable web applications. It follows the MVT (Model View Template) architecture. Django provides built-in features like authentication, admin panel, ORM, routing, and security.

Flask is another lightweight Python framework used for small web applications and APIs.

Python is considered one of the best programming languages for beginners because its syntax is very close to the English language.